



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI RAMAKRISHNAA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

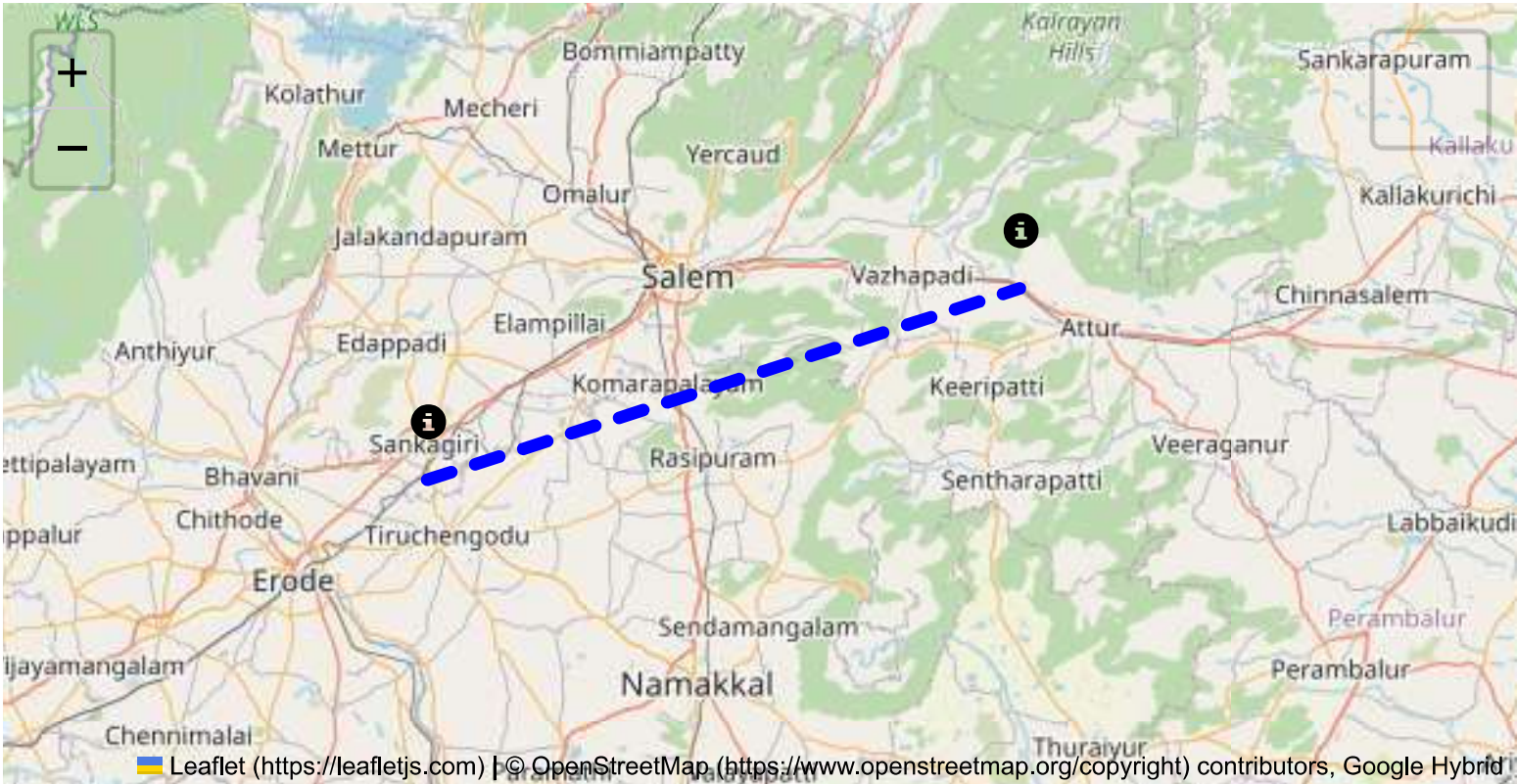
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

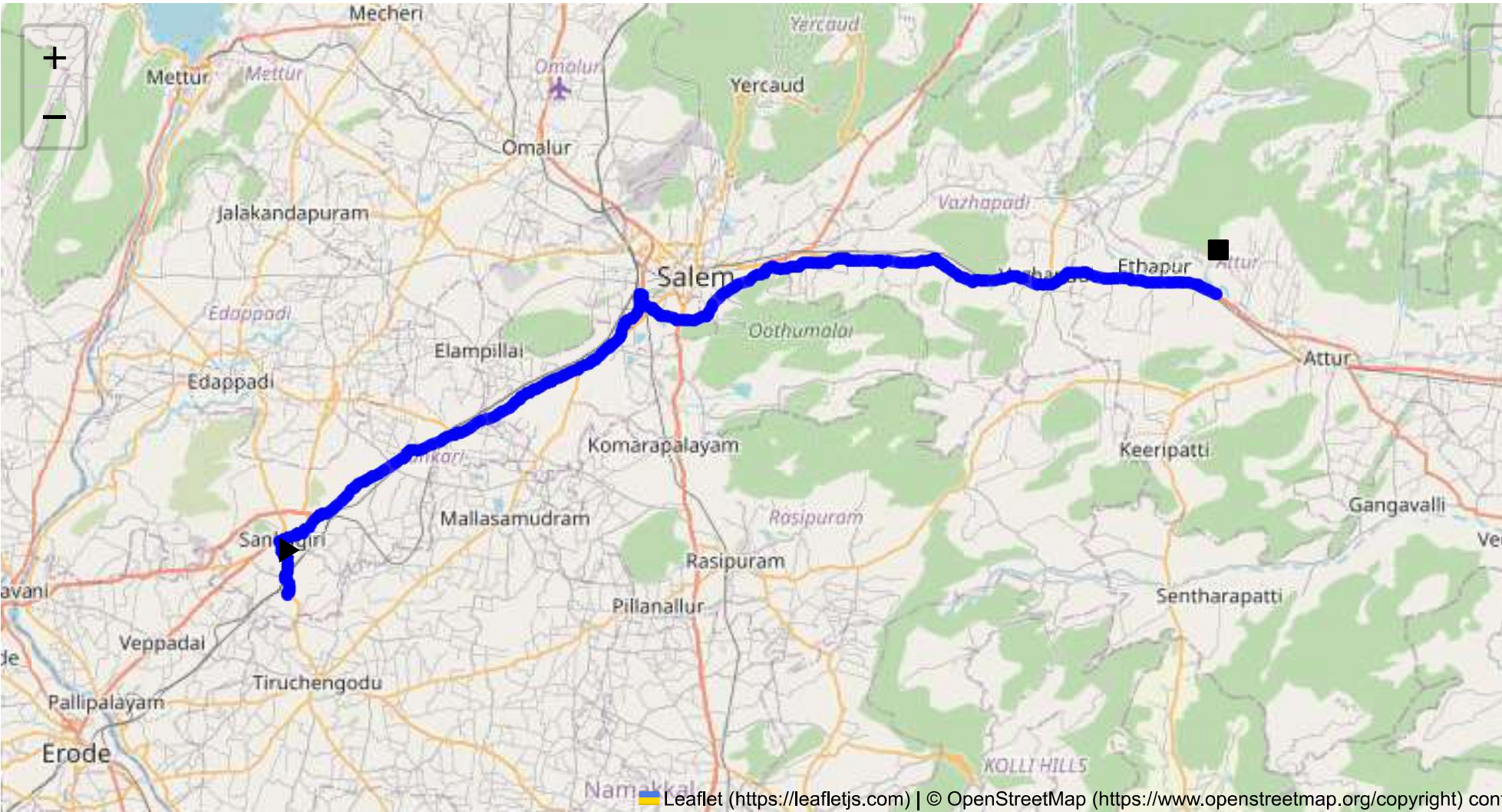
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 86.40 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (11.64264, 78.51947)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Pethanaickenpalayam is approximately 86.40 kilometers and generally takes around 1.65 hours for heavy vehicles. It primarily follows National Highway 79 (NH79), a major roadway facilitating trade and transportation across Tamil Nadu. This route is relatively straightforward, passing through a few smaller towns and rural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards: The region experiences a tropical climate, with heavy rainfall during the monsoon season (June to September). Storms can lead to waterlogged roads and reduced visibility. During the summer months (March to May), high temperatures can affect road surface conditions, potentially causing tire issues for heavy vehicles.

3. Analysis of Traffic Patterns: Traffic on NH79 can be moderately heavy, especially during peak hours (8-10 AM and 5-8 PM). Congestion is more prevalent near towns like Attur and areas where there are junctions or toll booths. The presence of slow-moving vehicles, including tractors and bullock carts, in rural areas can also impact travel time.

4. Assessment of Road Quality and Infrastructure: NH79 is generally well-maintained, with a dual carriageway providing separate lanes for opposing traffic. However, road quality can vary, especially near construction zones or areas of ongoing maintenance. Watch for potholes and uneven surfaces typically near village crossings.

5. Suggestions for Alternative Routes for Emergencies: In case of an emergency or severe congestion, an alternative route could be taking State Highway 86 (SH86) from Sangagiri to Attur, then reconnecting to NH79. This diversion offers a viable secondary path, albeit with slightly longer travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transporting hazardous materials in India requires compliance with the Motor Vehicles Act and relevant state regulations, which include vehicle safety standards, documentation, and specific permits. Restrictions may additionally apply during holidays and daytime hours in populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: While specific incident data may not be publicly available for this route, it's important to be aware of general risks such as heavy vehicle rollovers and spillage of hazardous materials. Ensuring adherence to safety protocols and vehicle checks can mitigate risk.

8. Environmental Considerations and Sensitive Areas: The route does not traverse any major ecologically sensitive zones; however, caution must be given to minimizing the environmental impact by following proper protocols for hazardous material transport. Awareness of fauna crossing regions is also advised, particularly in rural stretches.

9. Analysis of Communication Coverage: Mobile coverage is generally reliable along NH79, except in some remote sections between larger towns. It is advisable to ensure all communication devices are fully charged and to note key areas where network service may be sporadic.

10. Estimated Emergency Response Times for Different Route Segments: Emergency services in the larger towns along NH79 typically have response times ranging from 15 to 30 minutes. In rural areas or less populated stretches, response times can extend to 45 minutes to an hour due to distance and road conditions.

12. Overall Summary of Risk Assessment: This route offers a generally safe passage for transporting hazardous materials when adhering to traffic laws and safety protocols. Key risks include weather-related challenges, traffic congestion, and potential infrastructure issues. Ensuring vehicle safety and emergency preparedness is crucial in mitigating these risks, supported by local knowledge of peak traffic periods and alternative routes.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.61736, 78.10588	30 KM/Hr
13	Turn	Medium	11.61753, 78.10608	30 KM/Hr
14	Turn	Medium	11.64172, 78.11817	30 KM/Hr
15	Turn	Medium	11.64216, 78.11875	30 KM/Hr
16	Turn	Medium	11.64195, 78.12129	30 KM/Hr
17	Turn	Medium	11.64162, 78.12160	30 KM/Hr
18	Turn	Medium	11.64120, 78.12168	30 KM/Hr
19	Turn	Medium	11.63948, 78.12081	30 KM/Hr
20	Turn	Medium	11.65441, 78.38090	30 KM/Hr
21	Turn	High	11.64216, 78.51921	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
24	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
25	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
26	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
31	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
32	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
36	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
38	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
39	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
40	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
18	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
34	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
35	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
37	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64162, 78.12160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65441, 78.38090



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64216, 78.51921